

Task Description:

Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)

Install docker on EC2:

```
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1-24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `docker' (GID 113) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-35-81:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
ubuntu@ip-172-31-35-81:~$
```

i-0353ffa1239bbe1ea (Dockersys1)

PublicIPs: 65.0.96.212 PrivateIPs: 172.31.35.81

Docker Image Creation

```
root@ip-172-31-35-81:/home/ubuntu# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
taskimage     v1        222c2182ffc2   41 seconds ago 555MB
ubuntu        latest    c3a134f2ace4   5 weeks ago   78.1MB
root@ip-172-31-35-81:/home/ubuntu#
```

i-0353ffa1239bbe1ea (Dockersys1)

PublicIPs: 65.0.96.212 PrivateIPs: 172.31.35.81

Docker container Creation:

```
root@ip-172-31-35-81:/home/ubuntu# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
18ccd7bb8f0a   taskimage:v1   "python3 app.py"        39 seconds ago Exited (0) 39 seconds ago          unruffled_almeida
root@ip-172-31-35-81:/home/ubuntu# docker logs ^C
root@ip-172-31-35-81:/home/ubuntu# docker logs 18ccd7bb8f0a
Hello, Docker!
root@ip-172-31-35-81:/home/ubuntu#
```

Volume creation:

```
root@ip-172-31-35-81:/var/lib/docker/volumes# ls
backingFsBlockDev data metadata.db
root@ip-172-31-35-81:/var/lib/docker/volumes# cd data
root@ip-172-31-35-81:/var/lib/docker/volumes/data# ls
data
root@ip-172-31-35-81:/var/lib/docker/volumes/data# docker volumes ls
docker: unknown command: docker volumes

Run 'docker --help' for more information
root@ip-172-31-35-81:/var/lib/docker/volumes/data# docker volume ls
DRIVER      VOLUME NAME
local       data
```

Networks:

```
root@ip-172-31-35-81:/var/lib/docker/volumes/data# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
4d5ce6ce62c9        bridge             bridge             local
69efae0f7484        host               host               local
1bd3f572391e        none              null               local
```